

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Phase 2: Requirement Analysis & Technical Specification Document

PROJECT LEAD	DOCUMENT STATUS	TARGET PLATFORM	DATE
ServiceNow Systems Team	Approved Baseline	ServiceNow Utah / Vancouver	Q3 2026

1. BUSINESS REQUIREMENTS SPECIFICATION (BRS)

The requirement analysis phase translates the strategic vision established in Phase 1 into actionable, technical specifications. The objective is to implement a fully automated hardware procurement system within ServiceNow that reduces manual touchpoints from request to fulfillment.

Core Business Requirement Summary

Provide a unified, intuitive Service Catalog portal for users to request standardized hardware bundles while enforcing corporate budget policies, dynamic automated approvals, real-time inventory checks, and instant CMDB registration.

2. FUNCTIONAL REQUIREMENTS MATRIX

The following functional requirements have been gathered via stakeholder interviews across ITSM, Procurement, and Finance departments:

Req ID	Module / Domain	Functional Requirement Description	Priority (MoSCoW)
FR-01	Service Catalog	Single Service Catalog Item with dynamic variable cascading (User Location, Role Tier, Delivery Address validation).	Must Have
FR-02	Decision Engine	Flow Designer Decision Tables to dynamically route approvals based on requester manager hierarchy and cost center thresholds.	Must Have
FR-03	Inventory Management	Automated query to Hardware Asset Management (HAM) tables to verify local stock availability before triggering external vendor PO.	Must Have
FR-04	Vendor Integration	Outbound REST API call triggered via Flow Designer REST Spoke to dispatch order requests directly to authorized suppliers.	Should Have
FR-05	User Notifications	Integration with Microsoft Teams Spoke to send actionable cards for approvals and milestone delivery notifications.	Must Have

3. USER PERSONA & STAKEHOLDER IMPACT ANALYSIS

End User Simplified 3-click ordering experience with 100% status transparency via Teams.	IT Admin Elimination of 15+ hours/week of manual PO entry & manual CMDB updates.	Finance Manager Automated budget threshold enforcement with instant audit trail logs.
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4. TECHNICAL & NON-FUNCTIONAL REQUIREMENTS

To ensure robust operation and scalability within the enterprise environment, the platform architecture must comply with stringent technical standards across security, performance, and data architecture.

4.1 Non-Functional Requirements (NFRs)

- **System Performance:** Flow Designer execution time for approval routing and task generation must complete in under 3 seconds post-form submission.
- **Security & Access Control:** Access to catalog submission strictly governed via Role-Based Access Control (RBAC) utilizing user criteria (sn_request_user).
- **API Resilience:** Vendor API Spoke integration must include automated retry logic (3 attempts with exponential backoff) in case of endpoint timeout.
- **Auditability:** Complete execution history and variable state changes logged in sys_flow_context for compliance auditing.

5. DETAILED FLOW DESIGNER LOGIC & SUB-FLOW ARCHITECTURE

The solution relies on a modular flow architecture comprising one primary trigger flow and three specialized sub-flows to ensure maintainability and reusability.

Component	Input Trigger / Condition	Core Actions & Output Tasks
Main Flow: Laptop Order	Catalog Item Submitted [sc_req_item]	Initializes flow, fetches user profile data, calls Sub-flow 1 for approvals, routes based on outcome.
Sub-flow 1: Dynamic Approval	Order Amount & Tier Level	Evaluates Decision Table. Auto-approves Standard Tier within budget; sends MS Teams Actionable Card to Manager for Tier upgrades.
Sub-flow 2: Asset Provisioning	Approval Granted = True	Queries alm_hardware. If stock > 0, generates local staging task. If stock = 0, triggers REST Spoke to Vendor.
Sub-flow 3: CMDB Sync	Asset Dispatched / Shipped	Creates or updates CI record in cmdb_ci_computer, updates asset status to 'In Transit' and assigns serial number.

6. DATA MAPPING & INTEGRATION SCHEMA

Data consistency between ServiceNow catalog variables, CMDB tables, and external vendor APIs is enforced via standardized data mapping rules:

- **Requester ID Mapping:** sys_user.sys_id mapped to sc_req_item.requested_for and Vendor API RecipientID.
- **Cost Center Tracking:** sys_user.cost_center auto-populates financial allocation codes on the generated Purchase Order.
- **Hardware CI Tagging:** Vendor delivery Webhook updates cmdb_ci_computer.serial_number and state to Pending Allocation.

7. REQUIREMENTS TRACEABILITY MATRIX (RTM)

The RTM validates that every functional requirement mapped during analysis connects directly to an architecture component and testing criteria for Phase 3 (Build & Implementation).

Req ID	Business Objective	Flow Designer Component	Validation Test Case
FR-01	Zero-touch request creation	Catalog Item: cat_item_laptop	Verify catalog item auto-fills user profile & manager.
FR-02	Accelerated approval workflow	Decision Table: dt_laptop_approval	Test auto-approval for <\$1,500 standard builds.
FR-03	Prevent over-ordering hardware	Action: Check Local Stockroom	Simulate stock > 0 (allocate) vs stock = 0 (PO trigger).
FR-04	Vendor automation pipeline	REST Spoke: Vendor_Order_v2	Validate JSON payload delivery to external API endpoint.
FR-05	Real-time user updates	MS Teams Spoke: Send Card	Confirm user receives status card upon milestone change.

8. RISK ASSESSMENT & MITIGATION PLAN

Potential operational risks identified during analysis, along with corresponding technical mitigations:

- **Risk: Vendor API Downtime** — *Mitigation:* Fallback queuing mechanism using ServiceNow Data Lookup tables to retry payloads automatically.
- **Risk: Approval Bottlenecks** — *Mitigation:* Automated 48-hour timeout in Flow Designer that escalates pending approvals to delegate managers.
- **Risk: CMDB Duplicate Records** — *Mitigation:* Identification & Reconciliation Engine (IRE) rules enforced during automated asset creation.

9. SIGN-OFF & TRANSITION TO PHASE 3 (BUILD)

Phase 2 Gate Review Status: APPROVED

Requirements baseline has been formally reviewed and agreed upon by IT Operations, Enterprise Architecture, and Procurement leads. The project is ready to proceed to Phase 3: Flow Designer Build, Integration Configuration, and Testing.

Stakeholder Role	Representative Name	Approval Status	Date
IT Operations Lead	[Name Placeholder]	Approved	Q3 2026
Procurement Manager	[Name Placeholder]	Approved	Q3 2026
Lead ServiceNow Architect	[Name Placeholder]	Approved	Q3 2026